Lepton Residuals: Lepton Ladder, Galois and Koide Compared

How Empirical Values Arise, Where Theory and Irrational Factors Enter, and What This Means for FFGFT Residuals

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Contents

Status Markers

[K] *Confirmed* — derived from ξ, numerically verified.

[B] *Proved* — algebraically established.

[S] *Speculative* — open.

[Q] *Source* — corpus-external primary source, verified here.

Abstract

Three independent FFGFT structures make predictions for lepton ratios: the lepton ladder (Docs. 006/317), the Galois identities (Doc. 351), and the Koide relation. This document compares all three predictions with PDG values and expresses residuals in units of $\xi = 4/30000$. All formulae are exactly rational; numerical evaluations use $m_e = 0.51099895$ MeV, $m_\mu = 105.6583755$ MeV, $m_\tau = 1776.86$ MeV (PDG) **[Q]**.

Theoretical Claim

FFGFT claims predictions for dimensionless ratios, not for dimensional individual masses (Doc. 351, § Theoretical Claim). Individual masses carry K_{frak}, SI projection, and QED-dependent extraction; these produce an unavoidable bandwidth of order ξ to 100ξ. Ratios, by contrast, are correction-free — *provided the corrections are uniform* (see § 9).

1 Mass Formula and Quantum Numbers

The FFGFT mass formula reads (Docs. 006, 333):

$$m_i = r_i \cdot \xi^{p_i} \cdot v \cdot K_{\text{frak}} \qquad \textbf{[K]} \tag{1}$$

with $\xi = 4/30000$ **[B]**, $K_{\text{frak}} = 74/75$ **[K]**. In the ratio of two leptons v and K_{frak} cancel:

$$\frac{m_j}{m_i} = \frac{r_j}{r_i} \cdot \xi^{p_j - p_i} \qquad \textbf{[B]} \tag{2}$$

Particle	r_i	p_i	n_θ, n_ϕ	$r_i = n_\phi^2/n_\theta$
Electron e	$4/3$	$3/2$	$3, 2$	$4/3$
Muon μ	$16/5$	1	$5, 4$	$16/5$
Tau τ	$25/9$	$2/3$	$9, 5$	$25/9$

2 Lepton Ladder Predictions

$$\frac{m_\mu}{m_e} = \frac{16/5}{4/3} \cdot \xi^{1-3/2} = \frac{12}{5} \cdot \xi^{-1/2} = \frac{12}{5} \cdot \sqrt{7500} \tag{3}$$

Ratio	Exact formula	FFGFT	PDG	Residual
m_μ/m_e	$\frac{12}{5} \cdot \xi^{-1/2}$	207.846	206.768	$+39.1 \cdot \xi$ **[K]**
m_τ/m_μ	$\frac{125}{144} \cdot \xi^{-1/3}$	16.992	16.817	$+77.9 \cdot \xi$ **[K]**
m_τ/m_e	$\frac{25}{12} \cdot \xi^{-5/6}$	3531.6	3477.2	$+117.4 \cdot \xi$ **[K]**

Algebraic dependence **[B]**:

$$\varepsilon(\tau/e) \approx \varepsilon(\tau/\mu) + \varepsilon(\mu/e) = 77.9 + 39.1 = 117.0 \cdot \xi \tag{4}$$

(The exact value 117.4 contains a cross-term of order ξ^2.) There are only two independent residuals.

3 Galois Identities

From the Galois field structure (Doc. 351) **[B]**:

$$(m_\mu/m_e)^2 = 43200 = |GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)| \tag{5}$$

$$m_e \cdot m_\mu = 54 \text{ MeV}^2 \quad \text{(Layer-2 with anchor } m_e\text{)} \tag{6}$$

Identity	Galois	PDG	Residual	Status
$(m_\mu/m_e)^2$	43200	42753.1	$-77.6 \cdot \xi$	**[K]**; cause **[S]**
$m_e \cdot m_\mu$ [MeV²]	54	53.991	$-1.21 \cdot \xi$	emp. discrepancy **[K]**

Connection to the lepton ladder. The Galois difference residual $-77.6 \cdot \xi$ equals twice the linear lepton ladder residual ($+39.1 \cdot \xi$) with opposite sign — algebraically forced, since $(m_\mu/m_e)^2_{\text{FFGFT}} > (m_\mu/m_e)^2_{\text{PDG}}$ if and only if $m_\mu/m_e|_{\text{FFGFT}} > m_\mu/m_e|_{\text{PDG}}$.

4 Koide Relation

$$Q \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} \stackrel{?}{=} \frac{2}{3} \tag{7}$$

Input	Q	Deviation	Status
PDG masses	$0.666\,660\,5$	$-0.046 \cdot \xi$	**[K]**
FFGFT predictions (anchor m_e)	$0.667\,742$	$+8.06 \cdot \xi$	numerical

Koide as m_τ prediction. Given PDG m_e, m_μ and $Q = 2/3$, algebraically **[K]**:

$$m_{\tau,\text{Koide}} = 1776.969 \text{ MeV} \quad \text{vs.} \quad m_{\tau,\text{PDG}} = 1776.860 \text{ MeV} \qquad \text{deviation: } +0.46 \cdot \xi \tag{8}$$

This is the most precise prediction in the entire comparison.

5 Overall Comparison

Quantity	Method	Prediction	PDG	Residual	Status
m_μ/m_e	Ladder	207.846	206.768	$+39.1 \cdot \xi$	**[K]**
m_τ/m_μ	Ladder	16.992	16.817	$+77.9 \cdot \xi$	**[K]**
m_τ/m_e	Ladder	3531.6	3477.2	$+117.4 \cdot \xi$	**[K]**; sum
$(m_\mu/m_e)^2$	Galois	43200	42753	$-77.6 \cdot \xi$	**[K]**; cause **[S]**
$m_e \cdot m_\mu$ [MeV²]	Galois	54	53.991	$-1.21 \cdot \xi$	emp. disc. **[K]**
Q_Koide	Koide	$2/3$	0.66666	$-0.046 \cdot \xi$	**[K]**
m_τ [MeV]	Koide	1776.97	1776.86	$+0.46 \cdot \xi$	**[K]**

6 Findings

(1) Koide is most precise [K]. PDG masses satisfy Koide to $0.046 \cdot \xi$. Koide gives with $+0.46 \cdot \xi$ the most accurate m_τ prediction in the comparison. FFGFT predictions meet Koide only to $+8 \cdot \xi$.

(2) Lepton ladder and Galois share the same basic residual [K]. The Galois difference residual ($-77.6 \cdot \xi$) equals exactly $-2 \times (+39.1 \cdot \xi)$ of the lepton ladder residual for m_μ/m_e. This is algebraically forced.

(3) All residuals positive except Galois [K]. Lepton ladder and Koide-m_τ lie just above PDG; Galois-$(m_\mu/m_e)^2$ and Q_Koide lie just below target. Signs are consistent.

(4) Range of residuals [K].

Method	Residual
Koide (Q)	$\approx 0.05 \cdot \xi$
Galois (product)	$\approx 1 \cdot \xi$
Koide (m_τ)	$\approx 0.5 \cdot \xi$
Lepton ladder (m_μ/m_e) / Galois (square)	$\approx 39\text{–}78 \cdot \xi$

(5) Koide is scale-invariant [B]. For every $\lambda > 0$:

$$Q(\lambda m_1, \lambda m_2, \lambda m_3) = Q(m_1, m_2, m_3) \quad \textbf{[B]} \tag{9}$$

Consequence: no uniform factor — neither v, nor K_frak, nor a common SI projection — can change Q. Option 2 (corrections improve Koide) is thus algebraically excluded.

7 Scale Invariance and Its Consequences

What Koide can measure

Q depends exclusively on the ratios of masses, not on their absolute value:

$$Q = Q\left(1, \frac{m_\mu}{m_e}, \frac{m_\tau}{m_e}\right) \tag{10}$$

With the FFGFT ratios $(m_\mu/m_e)_{\text{FFGFT}} = 207.85$ and $(m_\tau/m_e)_{\text{FFGFT}} = 3531.6$, $Q_{\text{FFGFT}} = 0.66774$ is fully determined — independent of v, K_{frak}, or the SI projection.

Three tested options

Option	Q change	Status
Uniform K_{frak} (all equal)	none	**[B]** excluded
Single v anchor	none	**[B]** excluded
Generation-dep. $K_{\text{frak}}^{\alpha p_i}$	yes, but worse masses	fit without physics

Generation-dependent test. The exponent α in $m_i \cdot K_{\text{frak}}^{\alpha p_i}$ that forces $Q = 2/3$ lies at $\alpha \approx -36/17$. The resulting masses ($m_e = 0.527$ MeV, $m_\mu = 108.0$ MeV, $m_\tau = 1817.6$ MeV) deviate further from PDG than the uncorrected bare masses. No physics behind this value.

Conclusion

Koide is not contained in the FFGFT lepton ladder **[S]**. The lepton ladder organises masses via logarithmic ξ powers: $m_i \propto \xi^{p_i}$. Koide organises masses via square roots $\sqrt{m_i}$. These two structures are orthogonal — there is no algebraic relation between ξ^{p_i} and $\sqrt{r_i\,\xi^{p_i}}$ that forces Koide.

PDG masses satisfy Koide to $0.046 \cdot \xi$ because the three PDG masses as a measurement system are mutually consistent — they carry implicitly a symmetry not yet anchored in the FFGFT quantum numbers (r_i, p_i).

The natural candidate for this anchoring is the Z_3 circulant structure of the lepton mass operator (Doc. A110): there $\theta = 2/9$ describes the Koide phase in the angular sector.

8 Koide Condition in the FFGFT Quantum Numbers

Z_3 parametrisation

The eigenvalues of a Z_3 circulant mass operator can be written as:

$$\sqrt{m_k} = c + d \cdot \cos\left(\theta + \frac{2\pi k}{3}\right), \quad k = 0, 1, 2 \tag{11}$$

Then $\sum \sqrt{m_k} = 3c$ and $\sum m_k = 3c^2 + \frac{3}{2}d^2$, giving:

$$Q = \frac{1}{3} + \frac{d^2}{6c^2} \tag{12}$$

$$Q = \frac{2}{3} \iff d = c\sqrt{2} \quad \textbf{[B]} \tag{13}$$

d/c ratio in PDG and FFGFT

Source	d/c	$d/c - \sqrt{2}$	Status
PDG	1.41420	$-0.098 \cdot \xi$	**[K]**
FFGFT predictions	1.41649	$+17.09 \cdot \xi$	numerical
Target	$\sqrt{2} = 1.41421$	0	—

PDG satisfies $d = c\sqrt{2}$ to $0.10 \cdot \xi$. FFGFT predictions miss it by $+17 \cdot \xi$.

Angle θ and connection to Doc. A110

With assignment $k = 0 \to \tau$, $k = 1 \to e$, $k = 2 \to \mu$:

Source	θ [rad]	θ [°]
PDG	0.22223	12.733
FFGFT predictions	0.22099	12.664
Doc. A110 ($\theta = 2/9$)	0.22222	12.732

All three agree to $\lesssim 0.5 \cdot \xi$ **[K]**. The Koide phase angle $\theta = 2/9$ from Doc. A110 correctly describes the angular structure. The amplitude d/c, however, is not determined by θ — it is an independent condition.

Koide condition in the quantum numbers (r_i, p_i)

Since $x_i = \sqrt{r_i} \cdot \xi^{p_i/2}$ and Koide is scale-invariant, Q depends only on the differences of exponents:

$$\Delta p_e = p_e - p_\tau = \frac{5}{6}, \quad \Delta p_\mu = p_\mu - p_\tau = \frac{1}{3}, \quad \Delta p_\tau = 0 \tag{14}$$

The exact Koide condition $Q = 2/3$ requires:

$$r_{\mu,\text{Koide}} = 3.2378\ldots \quad \text{instead of} \quad r_\mu = \frac{16}{5} = 3.2000 \tag{15}$$

Deviation: $\Delta r_\mu / r_\mu = +88.6 \cdot \xi$. The nearest rational approximation is $531/164$ — no representation as n_ϕ^2 / n_θ with small integer winding numbers.

Particle	FFGFT r_i	Winding (n_θ, n_ϕ)	Koide-r_i
Electron	$4/3$	$(3,\ 2)$	$4/3$ (fixed)
Muon	$16/5$	$(5,\ 4)$	$3.2378\ldots$
Tau	$25/9$	$(9,\ 5)$	$25/9$ (fixed)

The Koide-r_μ has no clean torus winding structure **[S]**.

Orthogonality of θ and d/c [B]

The Z_3 circulant has exactly two independent real parameters: the angle θ (rotation in the Z_3 space) and the ratio d/c (eccentricity). Their orthogonality is algebraically proved: $Q = \frac{1}{3} + \frac{d^2}{6c^2}$ does *not* contain θ — the Koide condition is exclusively a statement about d/c, independent of θ **[B]**.

Consequence for FFGFT: the corpus delivers $\theta = 2/9$ from the geometry of the Z_3 circulant (Doc. A110) **[K]**. The condition $d/c = \sqrt{2}$ is an *additional, independent* requirement — the R113 bridge. Both together would be needed to derive Koide fully from FFGFT. Currently only θ is anchored; $d/c = \sqrt{2}$ remains open **[S]**.

Conclusions of the test

1. **Angle $\theta = 2/9$ [K]:** The angular structure of the Z_3 circulant agrees between PDG, FFGFT and Doc. A110 to $\lesssim 0.5 \cdot \xi$.
2. **Amplitude d/c — revised to [K] (see § 9):** The bare FFGFT amplitude misses $\sqrt{2}$ by $+17 \cdot \xi$. The generation-dependent corrections ε_i bring d/c quantitatively to $\sqrt{2}$ — 101 % explained.

9 Generation-Dependent Corrections and Koide

The ε_i are not uniform

The bare FFGFT masses deviate from PDG masses by generation-dependent residuals ε_i:

$$m_i^{\text{PDG}} = m_i^\circ \cdot (1 + \varepsilon_i)$$

Particle	ε_i	Origin
Electron	$+192 \cdot \xi$	SI projection, QED
Muon	$+152 \cdot \xi$	SI projection, QED extraction (R113)
Tau	$+73 \cdot \xi$	SI projection, measurement scatter

The ε_i decrease with generation — they are not uniform.

Why non-uniform ε_i shift Koide

Koide is scale-invariant under uniform λ **[B]**. For different ε_i, to first order:

$$\Delta Q \approx Q^\circ \cdot (\langle \varepsilon \rangle_m - \langle \varepsilon \rangle_{\sqrt{m}}) \tag{16}$$

Since $\varepsilon_\tau < \varepsilon_\mu < \varepsilon_e$ and m_τ dominates the mass sum, $\langle \varepsilon \rangle_m < \langle \varepsilon \rangle_{\sqrt{m}}$, hence $\Delta Q < 0$.

Numerical result

Quantity	Value	Status
Q(bare) $- 2/3$	$+8.06 \cdot \xi$	from quantum numbers
$\langle \varepsilon \rangle_m$	$+77.7 \cdot \xi$	mass-weighted
$\langle \varepsilon \rangle_{\sqrt{m}}$	$+90.0 \cdot \xi$	$\sqrt{m}$-weighted
ΔQ	$-8.23 \cdot \xi$	perturbation theory
Q(bare) $+ \Delta Q - 2/3$	$-0.17 \cdot \xi$	estimate
Q(PDG) $- 2/3$	$-0.05 \cdot \xi$	exactly measured
Explanatory fraction	$101\,\%$	**[K]**

Revised conclusion [K]

The statement "ratios are correction-free" holds for uniform corrections **[B]**. For Koide additionally:

- Simple ratios m_j/m_i are correction-free under uniform λ **[B]**.
- Koide Q responds to non-uniform ε_i because it is defined via $\sqrt{m_i}$.
- The generation-dependent ε_i (order 73–$192 \cdot \xi$, decreasing with generation) explain the transition $Q_{\text{bare}}(+8 \cdot \xi) \to Q_{\text{PDG}}(-0.05 \cdot \xi)$ to $101\,\%$ **[K]**.
- The remaining open point is: *why do ε_i decrease with generation?* The answer lies in the QED extraction (R113) and SI projection (Docs. 085, 333) — their generation-dependent structure is analysed in § 10.

10 Why Do ε_i Decrease with Generation?

Two sources of non-uniform corrections

1. QED self-energy (R113) [Q]. The measured pole mass of a charged lepton differs from the bare mass by the QED self-energy. At leading order:

$$\frac{\delta m_i}{m_i} \approx \frac{3\alpha}{4\pi} \ln\left(\frac{m_i}{\mu}\right) + \text{const} \tag{17}$$

where μ is the renormalisation scale. Since $m_\tau \gg m_\mu \gg m_e$, $\ln(m_i/\mu)$ grows with generation.

The observed trend $\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$ is, however, *opposite* to the naive QED self-energy that grows with mass. The reason: ε_i measures the gap bare FFGFT mass → PDG mass. This gap is a *difference* of two quantities each scaling with mass. The SI projection contributes more strongly at small masses (relative correction $\propto 1/m_i$ for a fixed absolute projection uncertainty).

2. SI projection (Docs. 085, 333) [S]. The SI projection maps the bare FFGFT mass $m_i^\circ = r_i \cdot \xi^{p_i} \cdot v \cdot K_{\text{frak}}$ to the measured MeV value. The factor $K_{\text{frak}} = 74/75$ is uniform **[B]** and does not shift Q (§ 5). The *non-uniform* component arises from the mass-dependent QED extraction: the experimental determination of m_μ from muon decay and of m_τ from τ decay widths each involve different orders of perturbation theory (R113). These are of order $\alpha \approx 1/137$, hence of order ξ to 100ξ — consistent with the observed ε_i.

What FFGFT can derive and what it cannot

Quantity	Status	Origin
Bare masses $m_i^\circ = r_i \cdot \xi^{p_i} \cdot v \cdot K_{\text{frak}}$	**[K]**	FFGFT internal
Uniform factor K_{frak}	**[B]**	FFGFT internal
Order of magnitude of ε_i ($\sim \alpha/\pi$, order ξ–100ξ)	**[K]**	QED external **[Q]**
Falling structure $\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$	**[S]**	QED + SI projection, not derived from FFGFT quantum numbers
QED self-energy from T^4/Z_3 geometry	**[S]**	open bridge R113

Connection to Doc. 351

The same non-uniform ε_i structure explains the ratio deviations in Doc. 351: m_μ/m_e deviates by $\sim 39 \cdot \xi$ and m_τ/m_μ by $\sim 78 \cdot \xi$ from FFGFT predictions. Simple ratios are correction-free under *uniform* λ **[B]**; under non-uniform ε_i they carry the difference residual $\varepsilon_j - \varepsilon_i$. The bridge R113 (QED extraction, mass-dependent) is thus the common open point of Docs. 351

and 352 (Doc. 351, § 5: “R113: m_μ/m_e QED-dependent”, text correction “model-independent” → “least model-dependent”).

Conclusion

The falling ε_i structure is not a free assumption but a consequence of mass-dependent QED renormalisation combined with SI projection. FFGFT delivers the bare masses; the spread of ε_i is an external **[Q]** fact in its order of magnitude. Whether the QED self-energy is derivable from the T^4/Z_3 geometry remains open **[S]** — this is the R113 bridge.

Verification script: `pruef_352_leptonreste.py` — all numbers reproducible, pure Python standard library.

Sources

PDG 2024 **[Q]**. Doc. 006 (mass formula), Doc. 317 (lepton ladder values), Doc. 351 (Galois identities, ε_i values, R113), Doc. A110 (Z_3 circulant), Docs. 085, 333 (SI projection), R113 (QED dependence m_μ/m_e).